> 🛠️ **2026.10 수정본 (v11)** — 원본: ch02_03_02_decisionTree_diabetes_colab.ipynb
>
> `diabetes.csv` 를 저장소의 `data/diabetes.csv` 에서 읽도록 경로만 고쳤어요. 코랩에서는 이 파일을 `data/` 폴더에 올려 주세요.

01. 기본- 결정트리(decision tree)

- date : 22/05

데이터 셋 다운로드

- UCI : https://www.kaggle.com/uciml/pima-indians-diabetes-database
- 피마 인디언 당뇨병 발병 예측

(가) decision tree는 classification(분류)와 regression(회귀) 문제에 널리 사용하는 모델이다.
(나) 스무고개 놀이의 질문과 비슷하다.

In [1]:
```
# 라이브러리 불러오기
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
```

Data Fields

구분	설명	
Pregnancies	임신	
Glucose	포도당	
BloodPressure	혈압	
SkinThickness	피부두께	
Insulin	인슐린	
BMI	BMI	111
Diabetes Pedigree Function	당뇨병혈통기능	
Age	나이	
Outcome	결과	

In [2]:
```
pima = pd.read_csv("data/diabetes.csv")  # 2026.10 수정: 데이터 폴더 경로
```

In [3]:
```
pima.columns
```

Out[3]:
```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insuli
n',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='str')
```

In [4]:
```
pima.head()
```

Out[4]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigr
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

Feature Selection

In [5]:
```
pima.columns
```

Out[5]:
```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insuli
n',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='str')
```

In [6]:
```
pima.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

In [7]:
```
pima.head(3)
```

Out[7]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigr
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	

In [8]:
```
pima.columns
```

Out[8]:
```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insuli
n',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='str')
```

In [9]:
```
# 데이터 셋 (feature와 target 변수로 나누기)
feature_cols = ['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness',
       'BMI', 'DiabetesPedigreeFunction', 'Age']
X = pima[feature_cols] # Features
y = pima.Outcome       # Target variable
```

데이터 나누기

In [10]:
```
# 데이터 셋 나누기
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, rar
```

In [11]:
```
print(X_test.columns)
print(X_train.columns)
print(y_train.shape)
```

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insuli
n',
       'BMI', 'DiabetesPedigreeFunction', 'Age'],
      dtype='str')
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insuli
n',
       'BMI', 'DiabetesPedigreeFunction', 'Age'],
      dtype='str')
(537,)
```

In [12]:
```
# 의사결정 트리 모델 생성 및 학습
model = DecisionTreeClassifier(max_depth=5, random_state=0).fit(X_train,y_tr

# 예측
y_pred = model.predict(X_test)
```

모델 평가

In [13]:
```
from sklearn import metrics
```

In [14]:
```
# Model Accuracy, 얼마나 정확한가? 정확도
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.7575757575757576
```

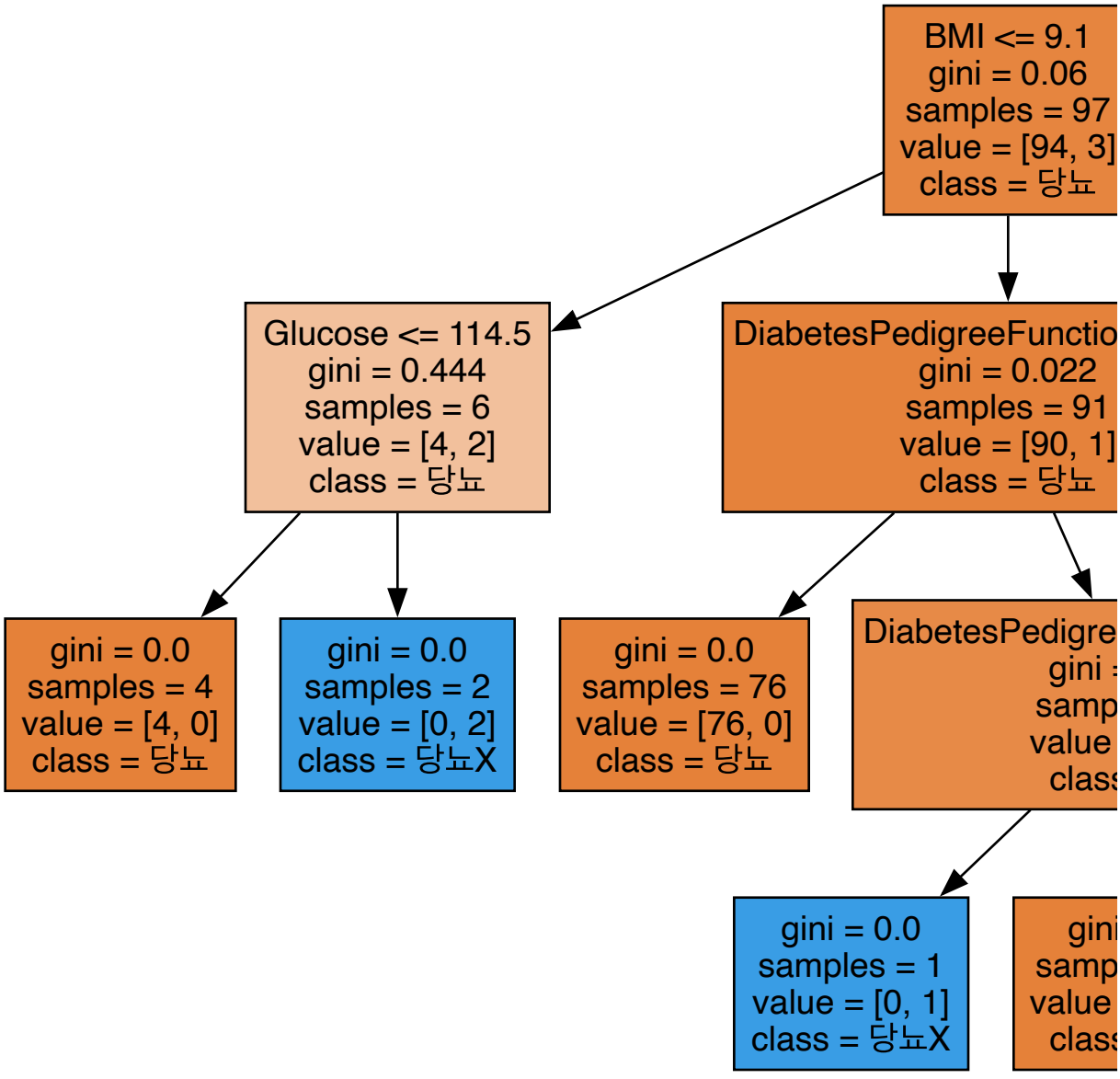
시각화 1

In [15]:

```
from sklearn.tree import export_graphviz
export_graphviz(model, out_file="tree.dot",
                class_names=['당뇨', '당뇨X'],
                feature_names = feature_cols,
                impurity = True,  # gini 계수
                filled=True)       # color

import graphviz
with open("tree.dot") as f:
  dot_graph = f.read()

display(graphviz.Source(dot_graph))
```


BMI <= 9.1
gini = 0.06
samples = 97
value = [94, 3]
class = 당뇨
Glucose <= 114.5
gini = 0.444
samples = 6
value = [4, 2]
class = 당뇨
DiabetesPedigreeFunctio
gini = 0.022
samples = 91
value = [90, 1]
class = 당뇨
gini = 0.0
samples = 4
value = [4, 0]
class = 당뇨
gini = 0.0
samples = 2
value = [0, 2]
class = 당뇨X
gini = 0.0
samples = 76
value = [76, 0]
class = 당뇨
DiabetesPedigre
gini =
samp
value
class
gini = 0.0
samples = 1
value = [0, 1]
class = 당뇨X
gini
samp
value
class

시각화 2

- png파일로 만들기

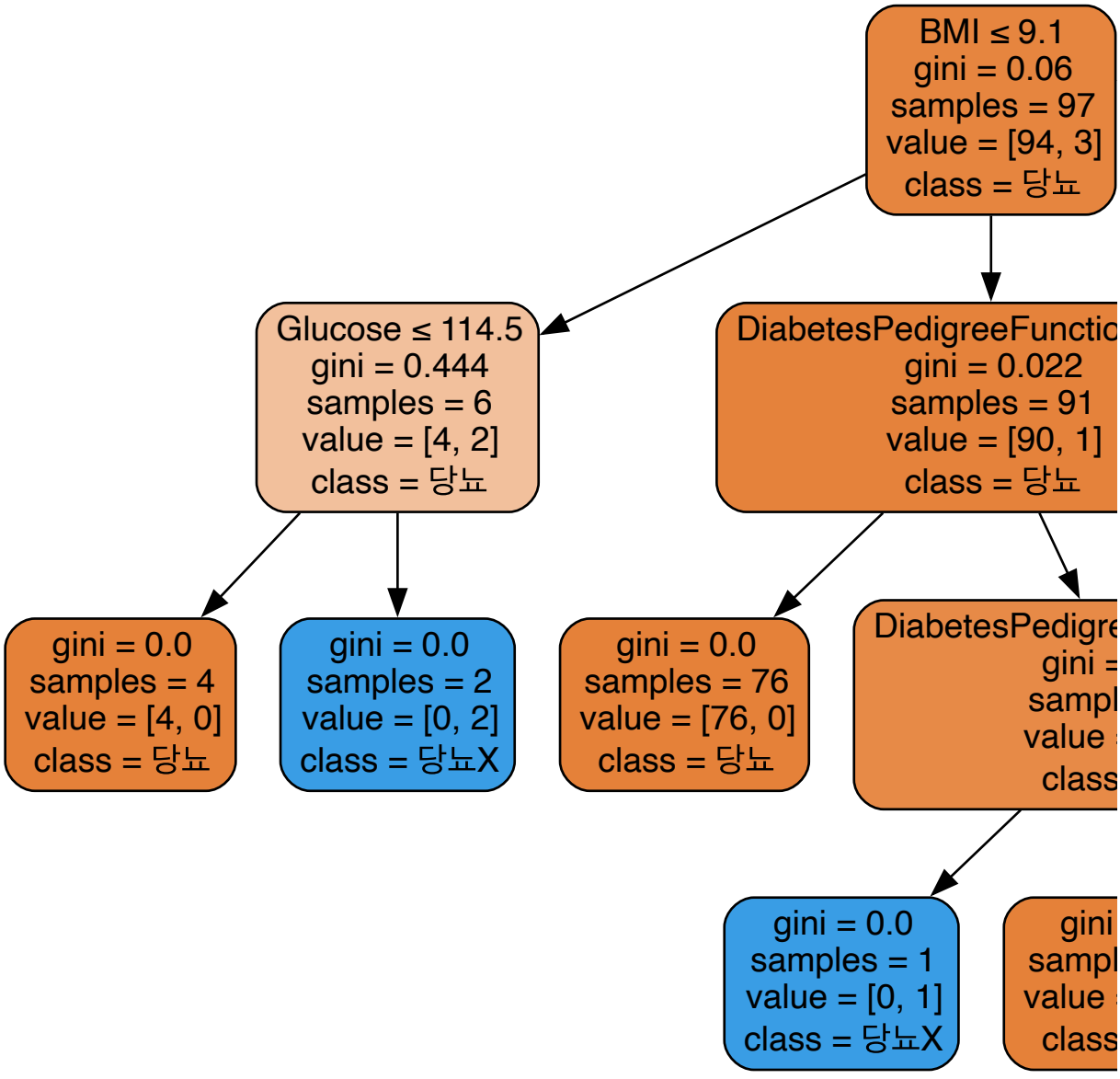
In [16]:
```
import pydotplus
from IPython.display import Image
```

In [17]:
```
import graphviz

# model : 모델명,
# class_n : 클래스명,
# feature_n : 특징 이름
def tree_plot(model, class_n, feature_n):
  export_graphviz(model, out_file="tree.dot",
                  class_names = class_n,
                  feature_names = feature_n,
                  impurity = True,  # gini 계수
                  filled=True,
                  rounded=True,
                  special_characters=True)       # color

  with open("tree.dot") as f:
    dot_graph = f.read()
  display(graphviz.Source(dot_graph))

tree_plot(model, ['당뇨', '당뇨X'], feature_cols)
```


BMI ≤ 9.1
gini = 0.06
samples = 97
value = [94, 3]
class = 당뇨
Glucose ≤ 114.5
gini = 0.444
samples = 6
value = [4, 2]
class = 당뇨
DiabetesPedigreeFunctio
gini = 0.022
samples = 91
value = [90, 1]
class = 당뇨
gini = 0.0
samples = 4
value = [4, 0]
class = 당뇨
gini = 0.0
samples = 2
value = [0, 2]
class = 당뇨X
gini = 0.0
samples = 76
value = [76, 0]
class = 당뇨
DiabetesPedigre
gini =
sampl
value
class
gini = 0.0
samples = 1
value = [0, 1]
class = 당뇨X
gini
sampl
value
class

모델 성능 개선

In [18]:
```
model = DecisionTreeClassifier(criterion="entropy",
                               max_depth=3,
                               random_state=0)  # 의사결정트리 모델
model.fit(X_train,y_train)   # 학습
y_pred = model.predict(X_test)  # 데이터 셋 예측

# 정확도 확인
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.7705627705627706
```